BIOMEDICAL LIMITED ILORIN	STANDARD TESTING PROCEDURE **TITLE:** STANDARD CHEMICAL TESTING PROCEDURE FOR BIOTHAZINE SYRUP (60ml)		
DEPARTMENT:	**QUALITY**	**SECTION:**	**QUALITY CONTROL**
STP NO.:	**BML/QA/STP/FPS/014/F01**	**PAGE NO.:**	**Page 1 of 5**
STP Ref.:	**BML-QDG-QA-105**	**REF. BOOK**	**BP 2024/IN-HOUSE**
EFFECTIVE DATE:	**04/01/2024**	**REVISION NO:**	**01**
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S/N	TEST	DESCRIPTIONS OF PROCEDURE
1.	**pH**	1. Take sample sufficient to immerse the electrode-head, and dip the electrode in the sample. 2. Wait until the reading stabilises, then take readings.
2.	**Promethazine HCl** **Label Claim: 5mg/5ml**	**METHOD 1** Reagents: a. DM b. Bromocresol Green solution (69.8mg of dye was dissolved in 2ml of 0.1N sodium Hydroxide and made to 1000ml with buffer solution pH 2.7) c. Buffer pH 2.7 (6g of Citric acid and 3.9g of disodium dihydrogen phosphate dissolved in about 900ml of water, pH adjusted to 2.7 – 2.8 with citric acid and diluted to 1000ml) d. Chloroform ***Standard solution***: 1mg/mL of Promethazine Hydrochloride in DM ***Sample solution:***

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		

		The sample containing 5mg/5ml of the substance was used as such. **Procedure:** To 2ml each of the sample and standard solutions, add 50ml of DM, 25ml dye solution. Mix and extract with three portions of chloroform. Pass the Chloroform layer through anhydrous sodium sulphate (already moistened with chloroform), Make up to 100ml. Measure the absorbance of both sample and standard at 415nm using chloroform as blank and calculate the results. **% Assay of Promethazine HCl** = $\frac{absorbance of sample}{absorbance of standard}$ x 100 ***REFERENCE:*** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 2
3.	**Microbial Enumeration Test**	- Determination of total aerobic microbial count (TAMC) and total combined yeasts and molds count (TYMC) by the pour plate method. - Dispense 1 ml of diluted sample in the center of the Petri plate using a sterile micropipette or calibrated pipette. - Open the lid of the prepared media, flame its mouth.

- Pour about 15 mL of sterilized molten media at the appropriate temperature above the sample.
- Close the lid of the plate then mix the sample in the media properly by gently swirling the plate. The plate is generally swirled in an “S” or “8” shape.
- Allow the media to completely solidify.
- Incubate the plate in an inverted position under suitable incubation conditions (Incubate the plates of Soybean Casein Digest Agar at 30°C-35°C for 3-5days for TAMC and the plates of Sabouraud Dextrose Agar at 20°C to 25°C for 5-7 days.

TEST FOR ABSENCE OF PATHOGEN

- Measure 10ml /10g of the samples, dilute 1n 90ml of nutrient broth
- Transfer 10ml into 90ml of Tryptone soya broth
- Incubate at 30-35^{0}c for 18-24hours
- Transfer 1 ml of the dilution into 100ml of MacConkey broth and incubate at 42-44°c in 18-72hours
- Subculture on MacConkey Agar or Eosin Methylene blue Agar and incubate at 30-35°c
- For negative control, proceeding same as above except without sample, streak a portion of incubated sterile MacConkey broth medium on the surface of Eosin

Methylene Blue Agar.

- Incubate plates at 30 to 35° for 48 hours.

A. Result Interpretation

- Following the incubation, developed colonies are observed. Each colony will account for one viable microbial cell or one colony-forming unit.
- If all the colonies are of the same type, then we can predict that the sample contained only one type of microbial genera.
- However, if there are different genera or species producing similar types of colonies, it is required to carry out identification test.
- If the colonies are of different morphology, then we can conclude that the sample contained a mixed population.

Count the colonies and calculate CFU/mL by using the formula:

$$\text{CFU/mL} = \frac{\text{Total number of colonies obtained x dilution factor}}{\text{Volume of sample used (aliquot)}}$$

- This will give the total number of viable microbial cells present in the given sample.
- For optimum count, the number of colonies must be between 20 – 200 CFU/mL. Beyond this limit, the whole procedure must be repeated.
- If the number of the colony is less than 20, it is suggested to use the sample of lower dilution, whereas, if the total number of colonies exceeds 200, it is suggested to use

		the sample of higher dilution on successive repeats. - The result should be recorded following the Standard testing procedure SOP **BML-QDG-QA-105** - If the colonies are fused or the whole plate is covered with a single colony, then report as “too numerous to count” (TNTC) and repeat the process of taking the sample at a higher dilution.

CURRENT REVISION	SUPERSEDE	REASON FOR REVIEW
00	NIL	FIRST ISSUE
01	00	Review of the procedures Change in header and footer format

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PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
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